No.	Paper	Year	Publisher	Main Novelty	Why it matches ThreatShield DL	Official Link
1	BALNet: Balanced Attack Learning Network for Web Attack Detection via Adaptive Sampling, LDA, and CNNs	2026	Springer Nature – *Discover Computing*	Adaptive Sampling + LDA + CNN for multi-class web attack detection	Directly aligns with your project's objective of deep learning-based web attack detection (SQLi, XSS, etc.) and serves as the research foundation.	https://link.springer.com/article/10.1007/s10791-026-10117-w
2	Swift Detection of XSS Attacks: Enhancing XSS Attack Detection by Leveraging Hybrid Semantic Embeddings and AI Techniques	2025	Springer	**Hybrid semantic embeddings** for XSS detection	Introduces NLP-based semantic embeddings for XSS detection, providing an alternative feature representation approach	https://link.springer.com/article/10.1007/s13369-024-09140-0
3	UniEmbed: A Novel Approach to Detect XSS and SQL Injection Attacks Leveraging Multiple Feature Fusion	2025	Springer	Multiple Feature Fusion (Word2Vec + USE + FastText)	Demonstrates advanced embedding fusion techniques for SQL Injection and XSS detection.	https://link.springer.com/article/10.1007/s13369-024-09916-4
4	Web-Based Phishing URL Detection Model Using Deep Learning Optimization Techniques	2025	Springer	Variational Autoencoder (VAE) + EGSO Optimization + CNN	Supports the phishing URL detection module and shows optimization-based feature extraction.	https://link.springer.com/article/10.1007/s41060-025-00728-9

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5	HawkPhish-DNN Cybersecurity Model: Adaptive Hybrid	2025	Springer	Harris Hawk Optimization + Deep Neural Network	Introduces metaheuristic optimization to improve phishing detection accuracy.	https://link.springer.com/article/10.1007/s41870-025-02597-8
6	Penetration Tests on Web Applications Using SQL Injection Attacks	2025	IEEE	**Experimental SQL Injection penetration testing**	Supports the SQL Injection module by demonstrating practical attack validation and security assessment.	https://ieeexplore.ieee.org/document/11304111
7	Phishing Behavior Detection on Different Blockchains via Adversarial Domain Adaptation	2024	Springer	**Adversarial domain adaptation** for phishing behavior	Presents domain adaptation rather than conventional classifiers	https://link.springer.com/article/10.1186/s42400-024-00237-5
8	L-xaids: A LIME-based eXplainable AI Framework for Intrusion Detection Systems	2025	Springer	Explainable AI (LIME-based IDS)	Supports the Explainable AI module of ThreatShield DL by providing interpretable attack predictions and improving user trust.	https://link.springer.com/article/10.1007/s10586-025-05326-9
9	BS-GAT: A Network Intrusion Detection System Based on Graph Neural Network for Edge Computing	2025	Springer	Graph Attention Network (GAT)	Introduces Graph Neural Networks for intrusion detection, offering a different deep learning architecture compared to CNN and BiLSTM.	https://link.springer.com/article/10.1186/s42400-024-00296-8

No.	Paper	Year	Publisher	Main Novelty	Why it matches ThreatShield DL	Official Link
10	Federated Learning for Intrusion Detection in IoT Environments: A Privacy-Preserving Strategy	2025	Springer	Federated Learning	Introduces privacy-preserving distributed learning, extending intrusion detection beyond centralized deep learning models.	https://link.springer.com/article/10.1007/s43926-025-00169-7